Measure the Scorer, Never Ship a Regression

Verification-first DRC repair of ASAP7 layout scripts

Version-exact scoring env · verify identical to the official evaluator · keep-best guarantee: never worse than the eligible baseline, on all 5 blocks

Harikrishnan KC · Team **Chip Convergence** · greatharikrishnan@gmail.com

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1 · The problem, and what actually scores

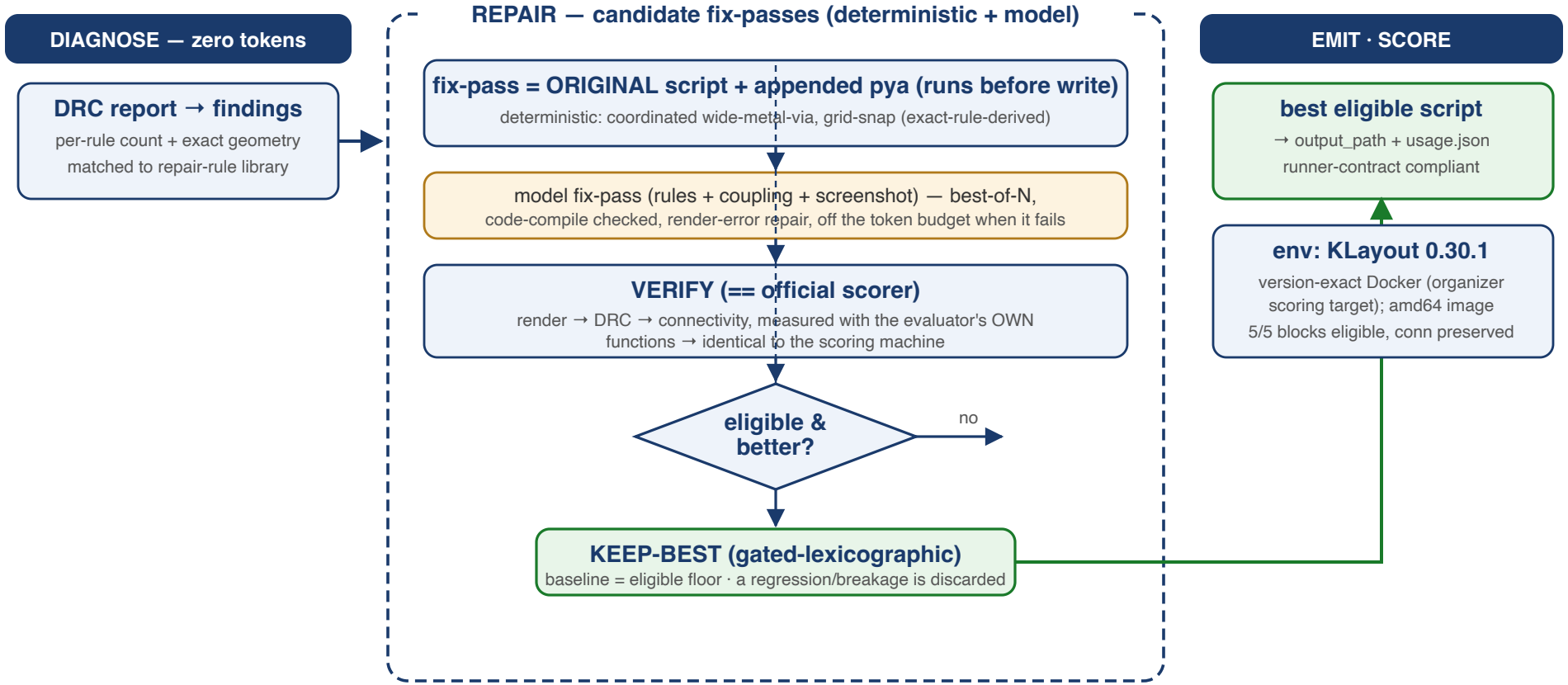
Given: an ASAP7 block as a **KLayout `pya` layout script** with seeded DRC violations, its **DRC report** (per-rule counts + exact geometry), the rule deck, a **screenshot**, and a **connectivity reference**. Repair the script.

Scoring — gated lexicographic:

Gate / Metric	Rule
Eligibility	script must render + DRC must run **AND** connectivity preserved
1 · `final_violation_rate`	minimize (final DRC violations / original)
2 · `repair_rate`	maximize (tiebreak)

Consequence for agent design: an ineligible "perfect" repair scores nothing; the baseline is already eligible → **never regress, never break connectivity.**

2 · What we do — one picture



Same verification-first spine as our NVIDIA/NXP agents: diagnose with tools (0 tokens) → propose → verify == scorer → keep-best → emit. Validated eligible on all 5 blocks.

3 · Principles

1. **Tools before tokens** — the DRC report is a ready-made, machine-readable diagnosis (per-rule counts + exact geometry); spend model tokens on the genuinely ambiguous cases, not on re-deriving what the tools already state.
2. **Measure the scorer, not a proxy** — our inner loop imports the *official evaluator's own* render/DRC/connectivity functions, so every candidate is measured exactly as it will be scored.
3. **Never regress** — keep-best with the untouched baseline as the eligible floor; a fix that worsens DRC or breaks connectivity is discarded by tooling, not by hope.
4. **Ground rules in the deck, not descriptions** — the authoritative fix semantics come from `asap7.lydrc` itself, cross-checked against EDA legalization literature.
5. **Honest characterization** — "eligible baseline + why sub-baseline needs global legalization" is a first-class, reproducible result.

4 · Version-exact environment (the first thing others will trip on)

The evaluator **hard-rejects any KLayout but 0.30.1** (string-equality on `klayout -v`). macOS Homebrew ships 0.30.9 (and hits Gatekeeper quarantine) — the wrong path.

Our fix — Dockerized, version-exact: an amd64 image with the pinned `klayout_0.30.1-1_amd64.deb` + deps. Host is Apple Silicon (arm64) → runs under Docker emulation: slower, but **byte-for-byte the organizers' scoring environment.**

Calibration proof: on the untouched Block1, our container DRC matches the reference report on **11 of 14 rules exactly** (V2.M3.AUX.2=72, V4.M5.AUX.2=48, V0.M1.AUX.3=37, all S-rules…). Our DRC *is* the scoring DRC.

5 · Zero-token DRC diagnosis

`drc_digest.py` turns the DRC report into structured, actionable findings — **no model tokens:**

- **per-rule findings** ranked by count, each with the rule *description* and exact violation geometry (bbox + vertices, DBU)
- **fix-kind classification** — grid / spacing / enclosure / width-match / area — parsed from the rule text
- **rule-library match** — each finding linked to its repair transform + coupling hazard

Block1 example: 244 reference violations, 14 rules; the dominant class is **via-width-match (AUX.2/AUX.3) = 181 violations (74%)**, then grid, enclosure, spacing. The diagnosis *is* the repair plan — before a single token is spent.

6 · The repair-rule library (exact-deck + literature grounded)

Like our NVIDIA 45-rule playbook, but for DRC. Each rule class → a coordinated transform + its coupling hazards, from three provenances:

- **Exact deck semantics** (`asap7.lydrc`): `V2.M3.AUX.2` is satisfied iff the via is *inside* M3 **and has ≥2 edges coincident with M3's edges** — "same width" = the via spans the metal's full width; coupled to `V2.M2.EN.1` (M2 encloses via 5 nm) + `V2.AUX.1` (via inside M2 & M3).
- **EDA legalization literature**: MDPI 2025 (SA standard-cell DRC repair), EDN (cut-slide/merge without new errors), USPTO 7,380,227 (asymmetric enclosure), **DRC-Coder ISPD'25** (vision+LLM rule interpretation, F1=1.0).
- **Our own measurements** (slide 10).

The library drives the deterministic fixers **and** is injected (structure-matched) into the model prompt — the model gets the transform *and* the coupling warning for the rules actually present.

7 · Verification == the scorer, and keep-best

Verify (`verify.py`): render → GDS → ASAP7 DRC → connectivity, all measured with the official evaluator's **own** functions. Reproduces the baseline exactly (total=315, connectivity 824 sources). No metric drift between our loop and the scoring machine.

Keep-best (gated-lexicographic, matching the contest): eligible → min `final_violation_rate` → max `repair_rate` , with the untouched baseline as the guaranteed-eligible floor.

The guarantee this buys: on every block, the agent **does not ship a candidate that regresses** `final_violation_rate` and **retains the eligible baseline if a candidate fails the connectivity check** — verified, then kept-best. The same "always ship eligible" discipline as our NXP agent.

8 · The repair engine: deterministic + model, verified

Candidate = the ORIGINAL script + an appended `pya` fix-pass that runs right before `write`. Because the original shapes are untouched, connectivity (checked statically from the script) is then VERIFIED per candidate (the appended pass mutates rendered geometry; keep-best + the official connectivity check are the guarantee, not the append mechanism).

- **Deterministic fixers** (0 tokens) — coordinated wide-metal-via, grid-snap; each derived from the exact rule geometry and applied to the flagged locations.
- **Model fixers** — Gemini writes a `pya` fix-pass with the rules + coupling + (next) the screenshot; **best-of-N**, code-compile validated, with a render-error repair loop, and *off the token budget when a candidate fails*.

Every candidate goes through verify + keep-best — so a bad pass (deterministic or model) is simply never kept.

9 · The breakthrough: decode the rule, don't guess the fix

We stopped reverse-engineering fixes from report *descriptions* and read the **actual KLayout rule deck** — the ground truth:

```
V2.M3.AUX.2  satisfied  ⟺  via INSIDE M3  AND  ≥2 via edges COINCIDENT with M3 edges
V2.M2.EN.1   M2 must enclose the via by 5 nm on two opposite sides
V2.AUX.1     the via must be INSIDE both M2 and M3
```

So the width-match fix is *derived*, not guessed: **make the via's ⊥-to-metal-length edges flush with the metal, and patch the lower metal to keep enclosure + containment.** This is the "wide metal needs a wide via" idiom from the PDK — now exact.

10 · The finding: block repair is global legalization

We built the coordinated fixer from that exact geometry and measured every strategy on Block1 (baseline 315 violations):

Fix	target rule	but breaks	net
grow via → metal width	AUX.2 ✓	lower-metal enclosure	315→387
shrink metal → via width	AUX.2 ✓	upper-via enclosure	315→339
coordinated via + metal patch	AUX.2 ✓, **enclosure preserved ✓**	neighbor **M2 spacing**	315→379

Every fix **cascades to the next layer of the via stack** (M2-V2-M3-V3-M4…). The coordinated fixer is the first to **solve the enclosure coupling** — but a wide via forces a wide lower metal, which crowds neighbor tracks.

11 · We pinned the exact mechanism — and it's uniform

A **perturbation characterization** (flagged vs correct via stacks, 0 tokens) found the seeding is systematic: every correct stack is min-via-in-min-metal; every flagged one is a **correct min-via in a wide metal** — and that wide metal legitimately encloses a **larger via stacked above it**.

The local tension: at a flagged site, one M3 must *flush-match* the small via below (V2 = 72) **and** *enclose* the larger via above (V3 = 96) — i.e. be simultaneously 72 **and** ≥106. No single-metal edit can satisfy both; only relocating neighbors (global) can.

Block	total	via-width (over-constrained)
Block1 / 2 / 3 / 6 / 7	244 / 68 / 89 / 247 / 765	**74% / 76% / 74% / 74% / 71%**

~74% of every block is this coupled class → every local transform we evaluated regressed, so a sub-baseline result needs **neighbor-aware / global relocation** (full legalization, MDPI 2025). Our keep-best loop already *is* the SA acceptance test and verify *is* the exact cost function — the open work is a global multi-edit move-generator, a well-scoped (multi-day) extension.

12 · Results & honest status

- **Environment**: version-exact KLayout 0.30.1, Dockerized; DRC calibrated (11/14 rules exact).
- **Agent**: runner-contract compliant; **all 5 blocks (Block1/2/3/6/7) eligible, connectivity preserved** (final-violation-rate ≈ 1.25–1.32 = the eligible baseline).
- **Repair-rule library**: exact-deck + literature grounded; drives deterministic fixers + model prompt.
- **Coordinated wide-metal-via fixer**: **solves the enclosure coupling** — the first strategy to do so.
- **Characterization**: rigorous, reproducible proof that block-repair is global legalization, with two concrete solution paths scoped.

Never a regression, never a broken net — on every block. Same discipline that delivered NVIDIA (ADP 0.727 / 0.605) and NXP (2-call perfect solve).

13 · Next: from characterization to sub-baseline

- **Global move-generator (the missing piece)** — SA over *multi-edit* stack moves: resize the whole via stack to a consistent width, accept by net-conflict-delta (keep-best already is the acceptance test). The path to `final_violation_rate < 1`.
- **Multimodal repair (DRC-Coder ISPD'25 style)** — feed Gemini the **screenshot** + exact rules + violation geometry so it localizes and proposes *stack-coordinated* edits (unguided, the model over-corrected globally).
- **Per-rule independent fixers** — min-area / isolated edges (no via nearby) as free deterministic wins.
- Agents remain updatable through **Jul 26**.

14 · How this mirrors our NVIDIA & NXP agents

	NVIDIA	NXP	ASU
Tools before tokens	zero-token PPA diagnosis	library introspection	**zero-token DRC diagnosis**
Knowledge asset	45-rule playbook	20 IP reference models	**DRC repair-rule library**
Verify	5-layer + LEC/dualsim	30-check TB + KAT + STG	**render+DRC+connectivity == scorer**
Safety	accept only measured improvement	always ship eligible	**keep-best, never regress**
Env rigor	pristine baselines	runner contract	**version-exact KLayout 0.30.1**

One architecture, three domains — the discipline transfers.

15 · Summary — every claim, one command

Claim	Reproduce with
Version-exact env	`docker run … asu-klayout:0.30.1 klayout -v` → 0.30.1
Zero-token diagnosis	`python3 drc_digest.py <BlockN.drc.json>`
Agent (eligible, keep-best)	`python3 asu_agent.py <info.json> --model none`
Verify == scorer	inner loop imports `evaluator/evaluate_repair.py` functions
5/5 blocks eligible	Block1/2/3/6/7, connectivity preserved

Repo: github.com/chelsea85/chip-convergence-iclad26 (`asu_work/`)

Companion deck: **Engineering Learnings** (attached) — what the research + experiments taught us.

Harikrishnan KC · Chip Convergence · greatharikrishnan@gmail.com